

ZICHUN YE

Shanghai China ◇ alchemist@sjtu.edu.cn ◇ [Homepage](#)

Education

Shanghai Jiao Tong University

B.Eng. in Computer Science and Engineering

Sep. 2020 - Jun. 2024

Shanghai, China

- GPA 90.77/100 (or 3.90/4.3).
- Achieved A on more than 30 courses, including all theoretical computer science courses (Discrete Mathematics, Probabilistic Theory, Computing Theory, etc).

Shanghai Jiao Tong University

MA.Eng. in Computer Science and Engineering

Sep. 2024 - Mar. 2027 (Expected)

Shanghai, China

- GPA 3.87/4.0.

Publications

Y. He, **Zichun. Ye**, and C. Zhang. Understanding memory-regret trade-off for streaming stochastic multi-armed bandits. In Proceedings of the ACM-SIAM Symposium on Discrete Algorithms (SODA 2025), pages 3450–3485, 2025.

Zichun. Ye, R. Wang, X. Liu, and S. Li. Near-optimal regret for efficient stochastic combinatorial semi-bandits. arXiv preprint arXiv:2508.06247, 2025a.

Zichun. Ye, C. Zhang, and J. Zhao. Tight gap-dependent memory-regret trade-off for single-pass streaming stochastic multi-armed bandits. In International Computing and Combinatorics Conference (COCOON 2025), pages 209–222, 2025b.

Zichun. Ye, R. Wang, X. Wang, X. Liu, S. Li, and M. Hajiesmaili. Unlearning offline stochastic multi-armed bandits. *arXiv preprint arXiv:2605.00638*, 2026.

Research Experience

Memory-Regret Trade-Off for Streaming Stochastic Multi-Armed Bandits

May. 2023 – Mar. 2025

Advisor: Prof. Chihao Zhang, SJTU

- Characterized the memory-regret trade-off for streaming stochastic multi-armed bandits by introducing a P-pass streaming model with limited arm memory.
- Designed two streaming algorithms based on best-arm retention when memory is large and best-arm identification when memory is small and analyzed their regret bounds respectively.
- Proved the first tight minimax regret lower bound via the technique of likelihood, revealing sharp transitions between small-memory and large-memory regimes.
- Extended the setting to the gap-dependent regret and established tight regret bounds in the single pass.

Near-Optimal Regret for Efficient Stochastic Combinatorial Semi-Bandits

Mar. 2025 – June. 2025

Advisor: Prof. Xutong Liu, UW, Tacoma, Prof. Shuai Li, SJTU

- Studied the problem of achieving optimal regret for stochastic combinatorial semi-bandits without the extra $\log T$ factor, while keeping the algorithm computationally efficient.
- Designed CMOSS, a MOSS-inspired algorithm which incorporated new confidence bound into CMAB work.
- Proved near-optimal instance-independent regret bound by devising new technical methods and validated by ablation experiments on multiple algorithms.

Unlearning for Offline Stochastic Multi-Armed Bandits

Oct. 2025 – Jan. 2026

Advisor: Prof. Xutong Liu, UW, Tacoma, Prof. Mengfan Xu, UMass, Prof. Mohammad Hajiesmaili, UMass

- Formalized unlearning for offline stochastic bandits under privacy constraints, measuring utility by post-unlearning decision quality.
- Designed adaptive unlearning algorithms that switch between Gaussian noise injection and rollback.
- Established performance guarantees and lower bounds under different data-sampling models.

Swap Regret Minimization / Game Theory

Mar. 2026 – Present

Advisor: Prof. Haipeng Luo, USC

- Studied linear swap regret minimization under bandit feedback over general convex action sets, without assuming an efficient oracle for the set of affine maps that preserve the action set.
- Developed an outer-approximation exponential-weights algorithm with SCRiBLE-style exploration, perturbing a fixed point of the mean map along output-covariance directions and verifying each candidate action's feasibility before sampling.
- Refined high-order stability analysis to obtain near-linear dependence on the number of actions and designed a multi-prior algorithm for comparator-adaptive regret in multiplayer games.

Additional Experience and Awards

Teaching Assistant	<i>Topics in Modern Algorithms</i>	Fall 2023, 2024
Advisor: Prof. Chihao Zhang		
- The topics include moment method, concentration inequalities, martingales and diffusion model. - Responsible for designing, assigning, and marking homework.		
Bosch Scholarship		Dec. 2025
Zhiyuan Honor Scholarship (Top 10%)		Dec. 2021, 2022, 2023
Huawei Scholarship		Oct. 2022
COSCO First Class Scholarship		Oct. 2021

Technical Skills

Programming Languages	Python, C++, L ^A T _E X
Language	TOEFL 5.5 (107+/120, Reading 6, Listening 5.5, Speaking 5, Writing 5.5)
	GRE (157 VR 170 QR 4.0 AW)